

AI Medical Report Assistant

Project Overview

AI Medical Report Assistant is an end-to-end healthcare AI application designed to analyze chest X-ray images, detect pneumonia using a deep learning model, and generate AI-assisted medical reports using Google Gemini LLM.

Objectives

- Detect pneumonia from chest X-ray images
- Generate confidence scores
- Produce AI-generated radiology reports
- Build an end-to-end AI application
- Integrate deep learning with LLMs
- Store prediction history using SQLite
- Deploy backend on Render

Technologies Used

Frontend: Streamlit
Backend: FastAPI
Deep Learning: TensorFlow / Keras
Transfer Learning: EfficientNetB0
LLM: Gemini API
Database: SQLite
Deployment: Streamlit

Dataset Information

Dataset Used: Chest X-ray Images (Pneumonia) Dataset from Kaggle.
Total Images: Approximately 5,856
Classes: NORMAL and PNEUMONIA

Deep Learning Model

EfficientNetB0 transfer learning model was used for pneumonia classification.
Image Size: 224x224
Optimizer: Adam
Loss Function: Binary Crossentropy

Frontend Features

- Chest X-ray upload
- Real-time prediction
- Confidence visualization
- AI-generated report display

Backend Features

- FastAPI backend APIs
- Image prediction endpoint
- Health check endpoint
- SQLite integration

Gemini LLM Integration

Google Gemini API generates professional AI-assisted medical reports including findings, impression, severity assessment, recommendations, and disclaimer.

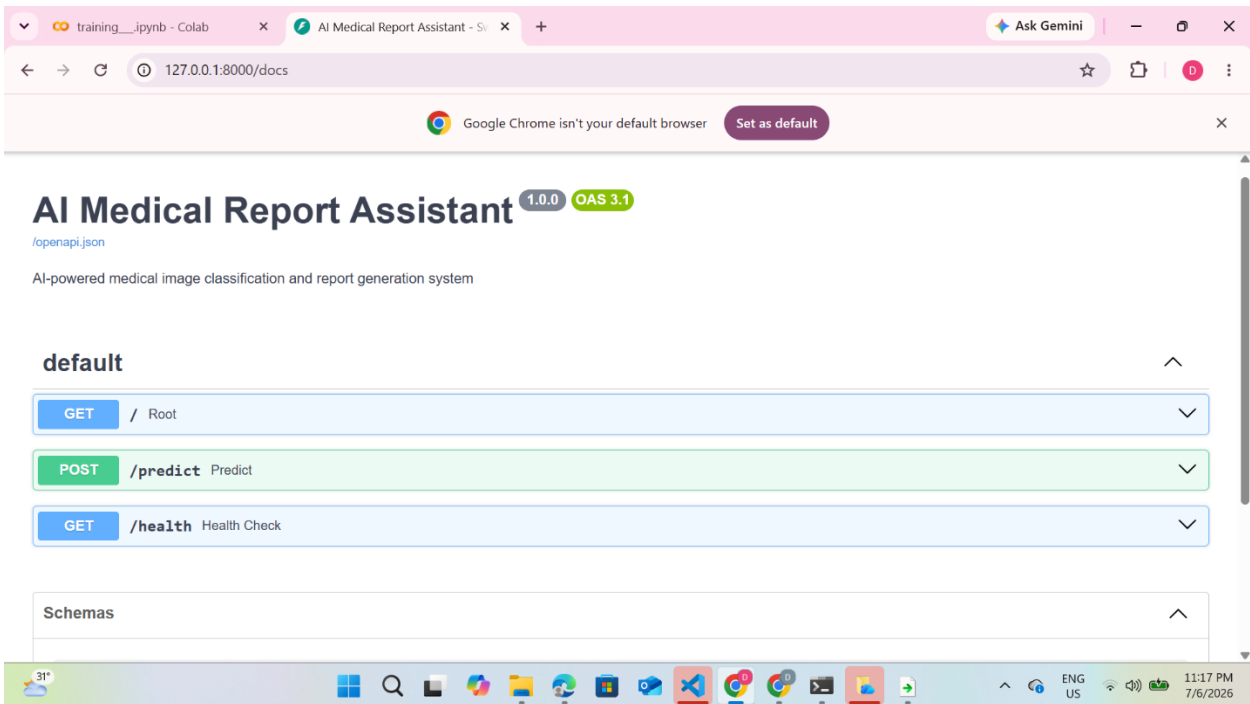
Deployment on Render

Backend is deployed on Streamlit cloud using FastAPI and Uvicorn.
Frontend runs using Streamlit.
Environment variables are configured using Render dashboard.

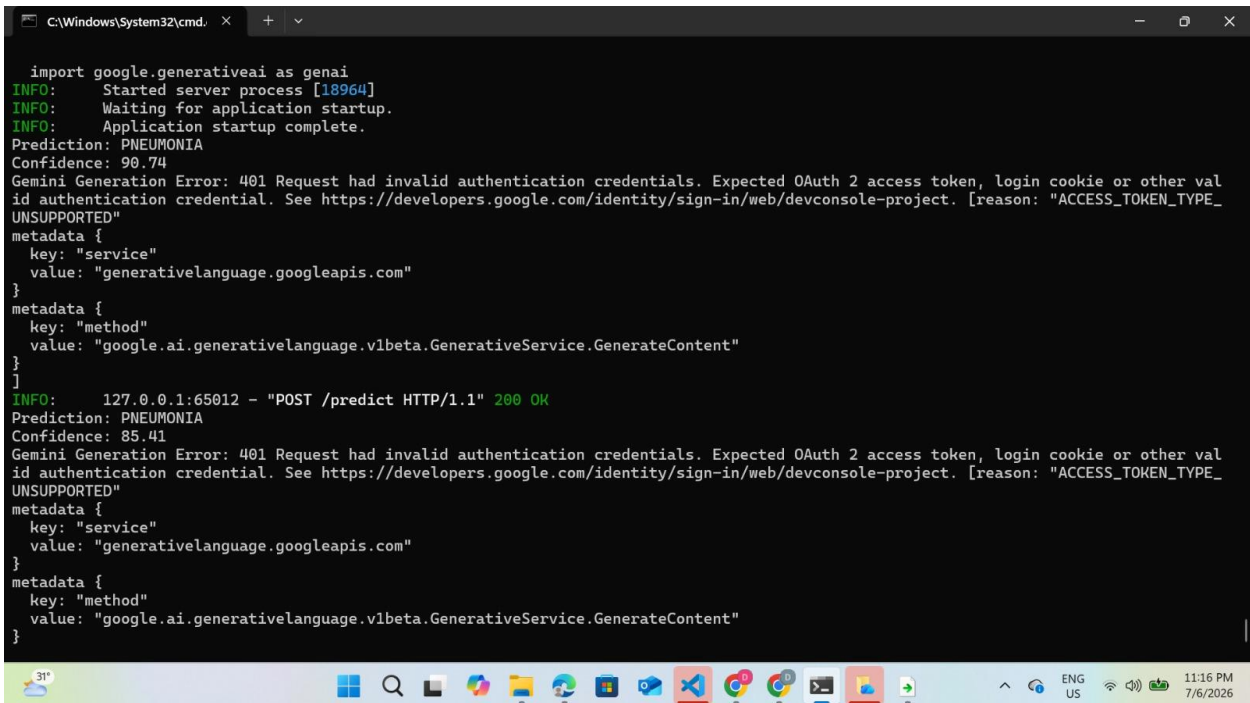
Requirements

```
tensorflow==2.16.1
keras==3.3.3
fastapi==0.111.0
streamlit==1.36.0
uvicorn==0.30.1
google-generativeai
python-dotenv
python-multipart
```

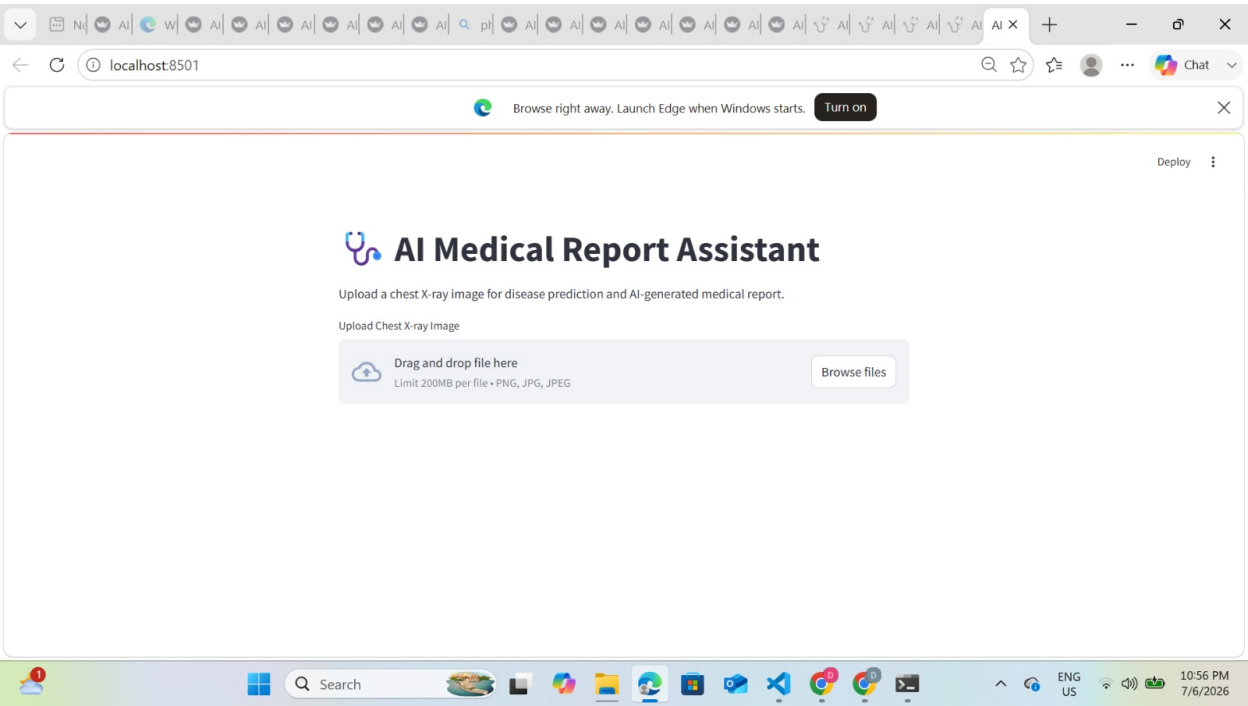
Backend API (Swagger UI)



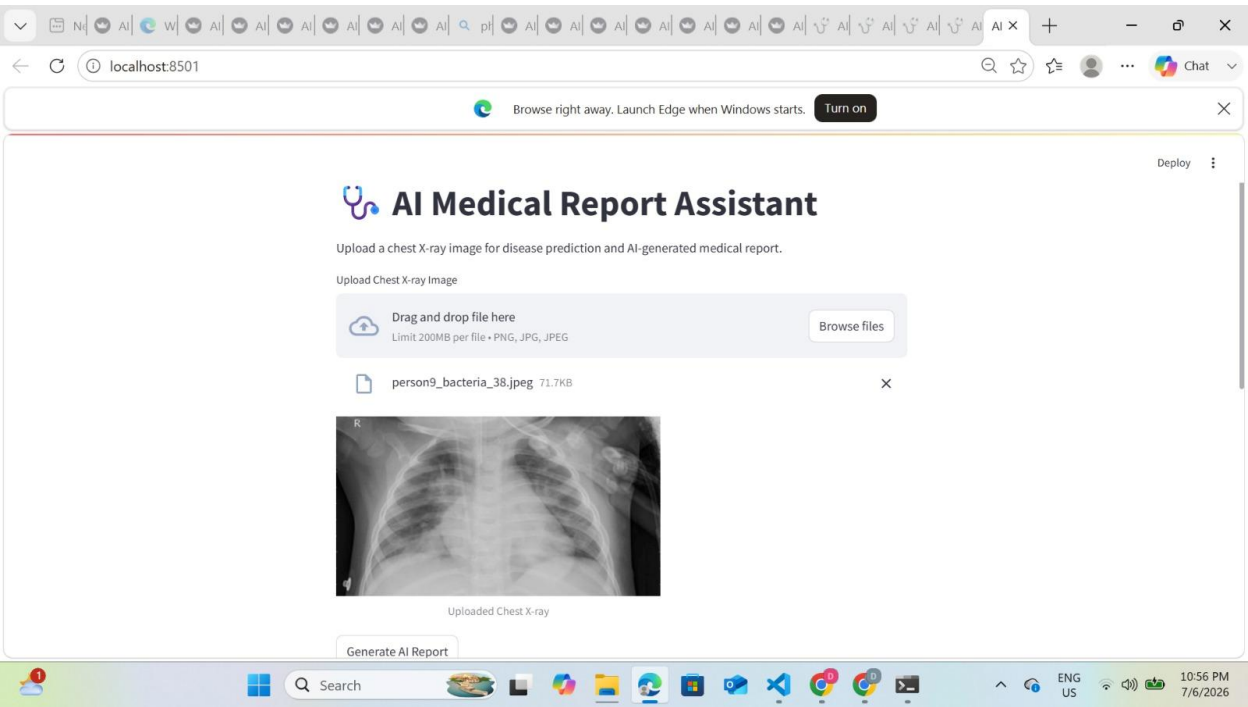
Backend Result

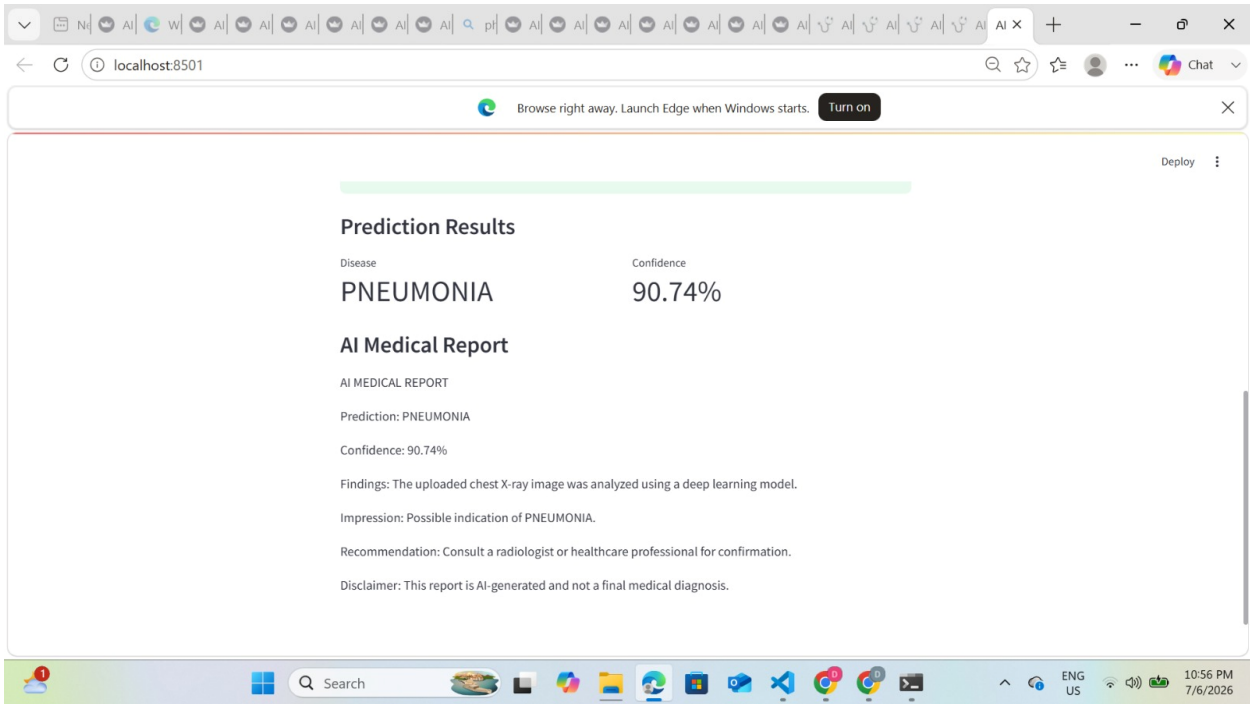


Prediction Page

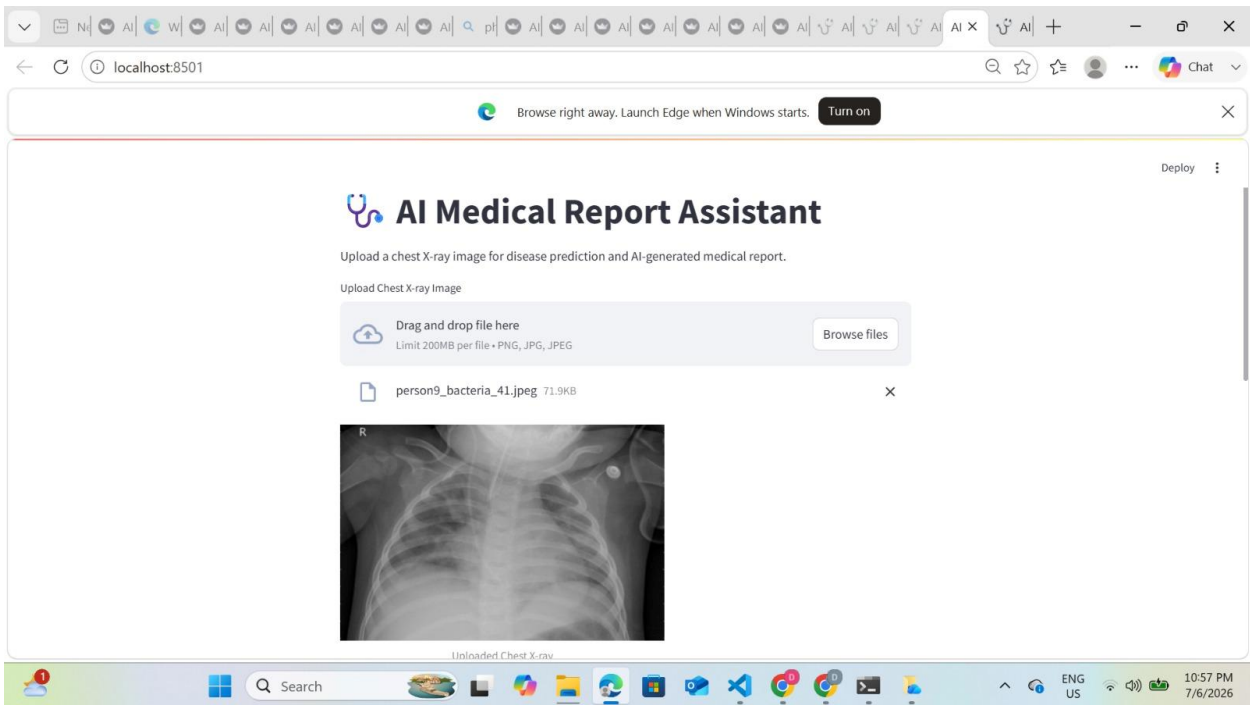


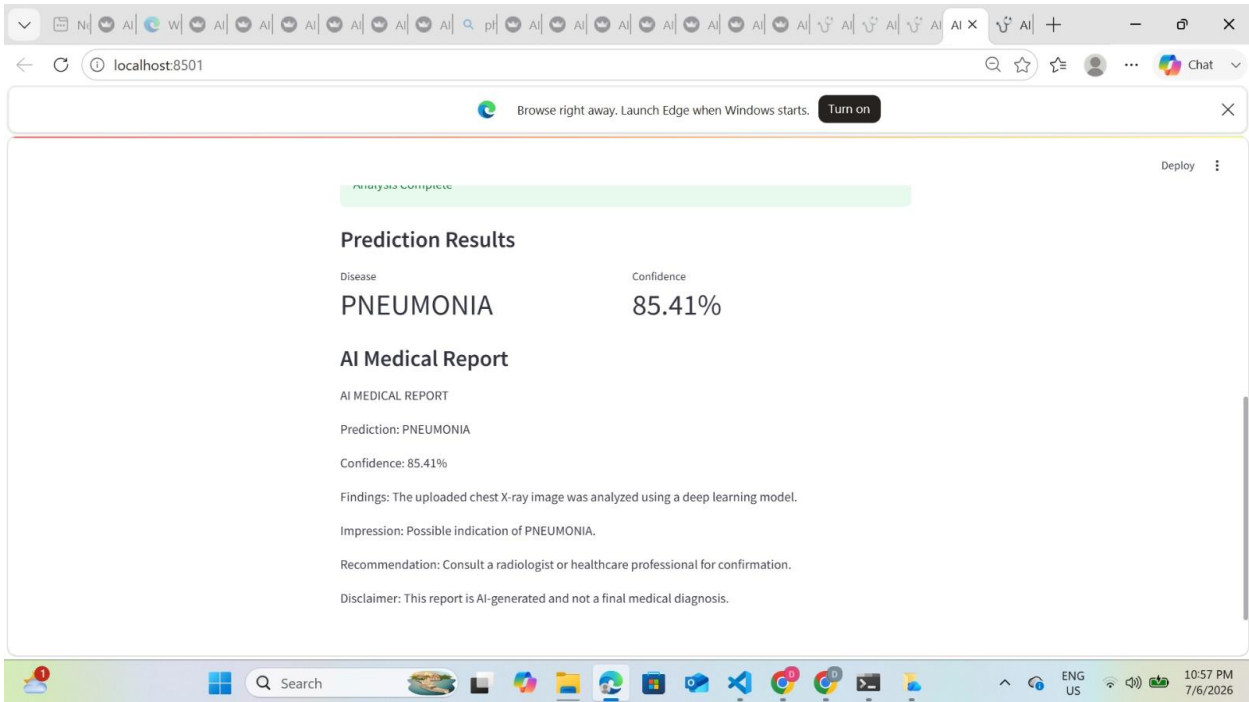
Prediction Result 1





Prediction Result 2





Conclusion

The AI Medical Report Assistant successfully demonstrates the integration of Deep Learning, Medical Image Analysis, FastAPI APIs, Streamlit applications, Gemini LLM, SQLite database, and cloud deployment.

Author

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